

COLLARDRIVE · DESIGN DECISIONS REGISTER

CollarDrive design decisions register

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DOCUMENT CONTROL

Document ID	CLD-DEC-001	Version	0.1
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/collardrive		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	Register opened; every decision made under Amish's 2026-10-03 pre-approval

CollarDrive design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list decisions.

Safety: Several decisions below are safety decisions (exhaust distance, backstop, guards, gas detector, no standing on the frame). Each takes the conservative option; the record says what evidence would relax it.

■ Open decisions

None. All decisions were made under Amish's 2026-10-03 pre-approval.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The pillow blocks' bolt centres and hole size (about 127 mm for UCP206 and 105 mm for UCP205)	They set the holes in the pedestal plates	CLD-DDR-002; CLD-DWG-103
2	The engine's foot pattern, shaft height (150 mm assumed) and muffler outlet size	They set the engine plate holes, the belt line and the exhaust hose	CLD-DDR-002; CLD-DWG-112
3	The blower's feet, outlet height (112 mm above its plate assumed), inlet flange and its curve at 2,000 rpm (0.15 m ³ /s at 500 Pa or better)	They set the blower plate holes, the flexible link and the air delivered	CLD-CAL-001, assumption 9
4	The V-belt lengths (about A126, A128 and A82) once the plates are set	Standard lengths are chosen from the measured centres	CLD-CAL-001 [A3]
5	The taper bush sizes for each pulley bought	They set the keyway positions on both shafts	CLD-DWG-106, CLD-DWG-107
6	The tyre sidewall thickness and the rope's grip in the V	Sets the wheel's bolt length and whether a liner is needed	CLD-DWG-108
7	The rising main's actual bore (36.2 mm assumed for 40 mm PN10)	Sets the piston diameter	CLD-DWG-120
8	The collar size and ground at the first site	Sleeper set-back and stake length	CLD-DDR-001, D7

■ Value engineering

Value-engineering target: USD 3,500 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 2,535 (USD 965 under the target). Main cost drivers and savings worth trying:

- The largest lines are the blower with its plate and inlet guard (USD 340), the gas detector (USD 320), the engine and plate (USD 215), 30 m of lay-flat duct (USD 195), the pulleys (USD 165) and the belt guard (USD 120).
- Savings worth trying: a locally made centrifugal blower in place of a bought one; a smaller 2 to 3 kW engine, since the load is under 0.8 kW (kept at the 4.8 kW class for now because it is sold and repaired everywhere); the

gas detector shared between kits at one site is not a saving to take, because every crew going below ground needs one.

■ Decisions made

Date	Decision	Decided by	Record
2026-10-03	Engine: 4.8 kW class petrol, governed to 2,800 rpm; a diesel of similar size fits the same plate	Amish: "I pre-approve the batch runs along with any recommendations you come up with."	CLD-DDR-001, D1
2026-10-03	Drive: engine to layshaft, two A belts to the pump, one A belt to the blower	Amish, same pre-approval	CLD-DDR-001, D2
2026-10-03	Idler clutch on the pump belts, lever latched in the guard	Amish, same pre-approval	CLD-DDR-001, D3
2026-10-03	Rope pump with a 400 mm tyre-sidewall wheel; main 40 mm to 30 m, 32 mm to 45 m, 25 mm to 60 m	Amish, same pre-approval	CLD-DDR-001, D4
2026-10-03	Forcing ventilation through 200 mm lay-flat duct	Amish, same pre-approval	CLD-DDR-001, D5
2026-10-03	Exhaust riser 2.4 m high, at least 6 m from the collar and the intake, downwind; the 6 m stays until intake carbon monoxide readings in varied winds show otherwise	Amish, same pre-approval	CLD-DDR-001, D6
2026-10-03	Two carried modules on three staked sleepers set at least 0.35 m back from the collar	Amish, same pre-approval	CLD-DDR-001, D7
2026-10-03	Ratchet backstop on every kit, until a test shows running back cannot hurt	Amish, same pre-approval	CLD-DDR-001, D8
2026-10-03	Full guarding of belts, pulleys, shafts and wheel; meshed blower inlet	Amish, same pre-approval	CLD-DDR-001, D9
2026-10-03	Personal carbon monoxide and oxygen detector in every kit	Amish, same pre-approval	CLD-DDR-001, D10
2026-10-03	Pump and air only; no hoist; the frame is never stood on; the collar is fenced	Amish, same pre-approval	CLD-DDR-001, D11
2026-10-03	Priced kit for a 30 m shaft with 30 m of duct	Amish, same pre-approval	CLD-DDR-001, D12
2026-10-03	First candidate partners to approach (not agreed): an artisanal mining cooperative through a national artisanal miners' body with a university mining department; a local rope pump maker through the Practica Foundation network	Amish, same pre-approval	CLD-DDR-001, D13

Date	Decision	Decided by	Record
2026-10-03	First candidate region to approach (not agreed): Zimbabwe (Mashonaland West), then Nigeria (Kano State)	Amish, same pre-approval	CLD-DDR-001, D14
2026-10-03	Design for construction: two bolted modules, sleepers and stakes, two-belt pump stage, idler clutch, pedestals, backstop, wheel, pipe head, guide block, duct head, slotted plates, guards, exhaust layout, discharge hose, crossbar positions	Amish, same pre-approval	CLD-DDR-002, Table 1
2026-10-03	Cost against the target: USD 2,535 against the USD 3,500 value-engineering target; budget_usd unchanged	Amish: "I also accept any cost overruns or variations from the assumed scope cost."	CLD-CAL-001 [H2]